NURICHIPS (FICTIONAL)

MX-7

13.56 MHz NFC/HF Reader IC

Datasheet v1.2
Silicon Rev A · 2025-03
Production Data

FICTIONAL DOCUMENT FOR TRAINING ONLY. Nurichips and the MX-7 do not exist. All data in this document is invented for an AI-training exercise and must not be used for any real design.

1 General Description

The MX-7 is a highly integrated 13.56 MHz NFC/HF reader IC for contactless communication in automotive, industrial and consumer applications. It supports ISO/IEC 14443 Type A and Type B reader mode and NFC Forum Type 2 Tag read/write operation. An on-chip transmitter drives the antenna directly, while the receiver demodulates and decodes card responses with integrated CRC and parity handling. The host communicates with the device through SPI at up to 10 MHz or I2C at 400 kHz, using a 64-byte FIFO for data exchange. A Low-Power Card Detection (LPCD) mode periodically senses the antenna field and wakes the host only when a card approaches. The device integrates a temperature sensor, operates from a single 2.7 V to 5.5 V supply over −40 °C to +125 °C ambient, and is offered in a QFN-32 5x5 mm package.

1.1 Features

- ISO/IEC 14443 Type A/B reader mode
- NFC Forum Type 2 Tag read/write
- Low-Power Card Detection (LPCD): 12 µA typ average current at the default 100 ms period
- 64-byte transmit/receive FIFO
- Host interface: SPI (up to 10 MHz) and I2C (400 kHz)
- Integrated temperature sensor, ±5 °C accuracy
- 27.12 MHz crystal oscillator
- Supply voltage VDD 2.7 V to 5.5 V (3.3 V typ)
- Operating ambient temperature −40 °C to +125 °C
- AEC-Q100 Grade 1 qualification in progress
- QFN-32 5x5 mm package

1.2 Applications

- Vehicle access and keyless entry readers
- Industrial access control and asset tracking
- Consumer NFC accessories and pairing
- Smart home and appliance authentication

Table 1-1. Key Specifications Summary

Parameter	Value
Carrier frequency (fc)	13.56 MHz
Supply voltage (VDD)	2.7–5.5 V (3.3 V typ)
Operating ambient temperature	−40 °C to +125 °C
Crystal	27.12 MHz ±50 ppm
Host interface	SPI up to 10 MHz / I2C 400 kHz
FIFO	64 bytes
RX active current	18 mA typ
Power-down current	1.0 µA typ / 3.0 µA max
LPCD average current	12 µA typ (100 ms default period)
Package	QFN-32 5x5 mm
Qualification	AEC-Q100 Grade 1 (qualification in progress)

Contents

1 General Description 1
2 Functional Overview 3
3 Pin Configuration 4
4 Absolute Maximum Ratings and Thermal Data 6
5 Electrical Specifications 7
6 Low-Power Card Detection (LPCD) 9
7 Protocol Timing 10
7.3 Frame Waiting Time (FWT) 10
8 Register Map 11
9 Command Set 12
10 Application Information 13
11 Package Information 14
12 Errata 15
Appendix A — Abbreviations 16
Revision History 16

Document Conventions

- Hexadecimal values are prefixed with 0x; binary values are suffixed with b.
- Unless otherwise noted, typical values are given at VDD = 3.3 V and TA = 25 °C.
- Register bits are written as NAME[msb:lsb]. Reserved bits read as 0 and must be written as 0.
- Access types: R = read only, R/W = read/write, R/W1C = read, write 1 to clear.
- fc denotes the 13.56 MHz carrier frequency.

2 Functional Overview

The MX-7 combines an RF analog front end, a protocol engine and a host interface in a single device. Figure 2-1 shows the main functional blocks. The transmitter generates the 13.56 MHz carrier derived from the 27.12 MHz crystal oscillator and modulates it according to the selected protocol. The receiver recovers the subcarrier from the antenna signal, and the framing unit performs bit decoding, parity checking and CRC verification before the payload is placed into the FIFO.

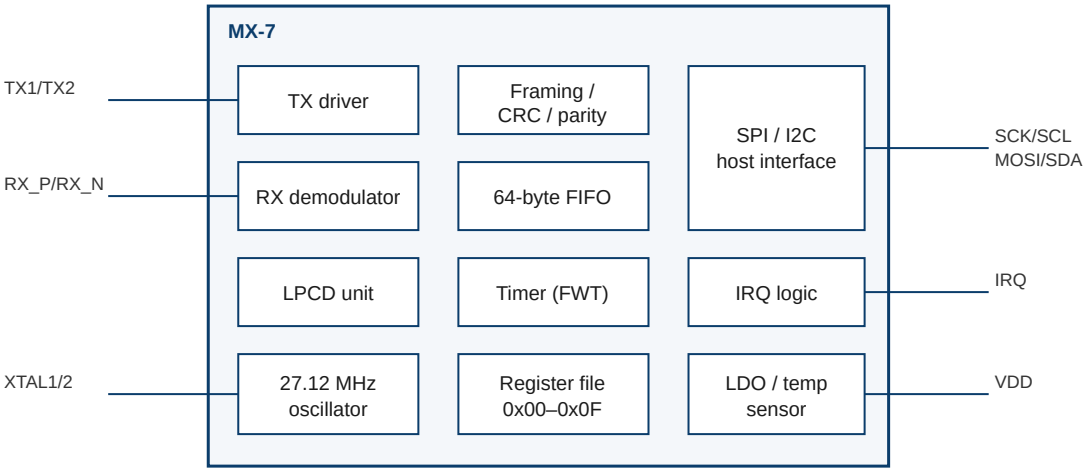


Figure 2-1. MX-7 Functional Block Diagram

2.1 RF Front End

The differential TX1/TX2 drivers connect to the antenna through an EMC low-pass filter and a matching network. The output drive strength is set by RF_CFG[3:2] (TX_DRIVE). The receiver uses the RX_P/RX_N inputs and a programmable gain stage controlled by RF_CFG[5:4] (RX_GAIN).

2.2 Protocol Engine

The protocol engine handles ISO/IEC 14443 Type A (Miller-coded 100% ASK) and Type B (NRZ-L, 10% ASK) reader mode framing at 106 kbit/s, with higher bit rates available for Type A. Hardware CRC generation and checking are controlled by TX_CFG and RX_CFG. NFC Forum Type 2 Tag READ and WRITE commands are executed with the TRANSCEIVE command (Section 9).

2.3 Timer Unit

A dedicated timer measures the Frame Waiting Time (FWT) after each transmitted frame. The FWT is configured by FWI_CFG[3:0] as described in Section 7.3. If no response is received within FWT, the timeout interrupt is raised in IRQ_STATUS.

2.4 FIFO Buffer

The MX-7 provides a 64-byte FIFO that is shared between transmit and receive directions. The FIFO is accessed through FIFO_DATA (0x03); the number of bytes currently stored is reported by FIFO_LEVEL (0x04), ranging from 0 to 64. Frames longer than the FIFO size must be serviced by the host using the FIFO level interrupts.

2.5 Power Modes

The device supports Active, Standby, LPCD and Power-down modes. In Power-down mode, all internal blocks except the reset logic are switched off and the supply current drops to 1.0 µA typ. The LPCD mode is described in Section 6.

3 Pin Configuration

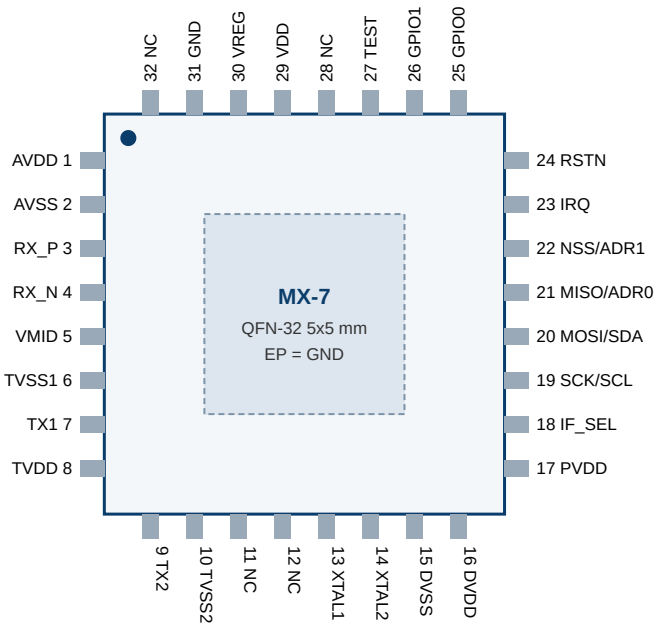


Figure 3-1. QFN-32 Pin Assignment (top view)

Pin type legend: P = power, G = ground, AI/AO = analog input/output, DI/DO = digital input/output, DIO = digital bidirectional.

Table 3-1. Pin Descriptions

Pin	Name	Type	Description
1	AVDD	P	Analog supply (internally regulated, decouple with 100 nF to AVSS)
2	AVSS	G	Analog ground
3	RX_P	AI	Receiver input, positive
4	RX_N	AI	Receiver input, negative
5	VMID	AO	Receiver reference voltage, decouple with 100 nF to AVSS
6	TVSS1	G	Transmitter ground 1
7	TX1	AO	Antenna driver output 1
8	TVDD	P	Transmitter supply, connect to VDD
9	TX2	AO	Antenna driver output 2
10	TVSS2	G	Transmitter ground 2
11	NC	–	Not connected; leave open
12	NC	–	Not connected; leave open
13	XTAL1	AI	27.12 MHz crystal input
14	XTAL2	AO	27.12 MHz crystal output
15	DVSS	G	Digital ground
16	DVDD	P	Digital core supply (internal LDO output), decouple with 100 nF
17	PVDD	P	Host I/O supply, 1.65 V to VDD
18	IF_SEL	DI	Host interface select: low = SPI, high = I2C (sampled at reset)
19	SCK/SCL	DI	SPI clock / I2C clock
20	MOSI/SDA	DIO	SPI data in / I2C data (open-drain)

Pin	Name	Type	Description
21	MISO/ADR0	DIO	SPI data out / I2C address bit 0
22	NSS/ADR1	DI	SPI chip select (active low) / I2C address bit 1
23	IRQ	DO	Interrupt request output, active low, push-pull
24	RSTN	DI	Hardware reset, active low, internal pull-up
25	GPIO0	DIO	General-purpose I/O 0
26	GPIO1	DIO	General-purpose I/O 1
27	TEST	DI	Factory test; connect to DVSS
28	NC	–	Not connected; leave open
29	VDD	P	Main supply input, 2.7 V to 5.5 V
30	VREG	AO	Internal regulator output, decouple with 1 µF
31	GND	G	Ground
32	NC	–	Not connected; leave open
EP	GND	G	Exposed pad; connect to PCB ground plane

4 Absolute Maximum Ratings and Thermal Data

Stresses beyond those listed in Table 4-1 may cause permanent damage to the device. These are stress ratings only; functional operation at these or any other conditions beyond those in Table 5-1 is not implied.

Table 4-1. Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Unit
Main supply voltage	VDD	−0.3	6.0	V
Host I/O supply voltage	PVDD	−0.3	6.0	V
Input voltage on digital pins	VIN	−0.3	PVDD + 0.3	V
TVDD supply current	ITVDD	—	250	mA
Junction temperature	Tj	—	150	°C
Storage temperature	TSTG	−55	150	°C
ESD, human body model (all pins)	VESD_HBM	—	±2000	V
ESD, charged device model (all pins)	VESD_CDM	—	±500	V

Table 4-2. Thermal Characteristics

Parameter	Symbol	Value	Unit
Junction-to-ambient thermal resistance (JEDEC 4-layer board)	θJA	38	°C/W
Junction-to-case (top) thermal resistance	θJC	12	°C/W
Junction-to-board thermal resistance	θJB	15	°C/W

The junction temperature can be estimated as Tj = TA + θJA × PD, where PD is the device power dissipation. With the RF field enabled, most of the dissipation occurs in the transmitter; designers operating at high ambient temperature should verify Tj under worst-case field-on duty cycle.

Table 4-3. Moisture Sensitivity and Reflow

Parameter	Value
Moisture sensitivity level (J-STD-020)	MSL 3
Peak reflow temperature	260 °C
Lead finish	Matte Sn
RoHS / halogen-free	Compliant

5 Electrical Specifications

5.1 Recommended Operating Conditions

Table 5-1. Recommended Operating Conditions

Parameter	Symbol	Min	Typ	Max	Unit
VDD supply voltage	VDD	2.7	3.3	5.5	V
Host I/O supply voltage	PVDD	1.65	3.3	VDD	V
Operating ambient temperature	TA	−40	—	+125	°C
Crystal frequency	fXTAL	—	27.12	—	MHz
Crystal frequency tolerance	Δf/f	−50	—	+50	ppm
Carrier frequency	fc	—	13.56	—	MHz

> Operating ambient temperature range: −40 °C to +125 °C. Crystal: 27.12 MHz ±50 ppm including temperature drift and aging. The carrier frequency fc = 13.56 MHz is derived by dividing the crystal frequency by two.

5.2 Electrical Characteristics

Table 5-2. Electrical Characteristics (VDD = 3.3 V, TA = 25 °C unless otherwise noted)

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
RX active current	IDD_RX	Receiver on, RF field off	—	18	—	mA
RF field-on current	IDD_RF	TX_DRIVE = 11b, reference antenna	—	120	—	mA
Standby current	IDD_STBY	Oscillator on, RF off	—	4.5	—	mA
Power-down current	IPD	RSTN low or power-down mode	—	1.0	3.0	µA
LPCD average current	ILPCD	Default period (100 ms)	—	12	—	µA
LPCD detection distance	dLPCD	40×40 mm reference antenna, LPCD_THR = 0x20	—	35	—	mm
Input high voltage (digital)	VIH		0.7 × PVDD	—	—	V
Input low voltage (digital)	VIL		—	—	0.3 × PVDD	V
Output high voltage	VOH	IOH = −4 mA	PVDD − 0.4	—	—	V
Output low voltage	VOL	IOL = 4 mA	—	—	0.4	V
Power-on reset threshold	VPOR	VDD rising	2.2	2.4	2.6	V

Summary at VDD = 3.3 V, 25 °C: RX active current 18 mA typ; RF field-on current 120 mA typ; power-down current 1.0 µA typ / 3.0 µA max; LPCD average current 12 µA typ at the default 100 ms period; LPCD detection distance 35 mm typ with the 40×40 mm reference antenna.

5.3 RF Characteristics

Table 5-3. RF Characteristics (VDD = 3.3 V, TA = 25 °C)

Parameter	Conditions	Min	Typ	Max	Unit
Carrier frequency fc	Crystal 27.12 MHz	—	13.56	—	MHz
Type A modulation depth	ASK	—	100	—	%
Type B modulation index	ASK	8	10	14	%
Supported bit rates, Type A		106	—	848	kbit/s
Supported bit rate, Type B		—	106	—	kbit/s
Minimum RX input signal	Subcarrier 847.5 kHz	—	1.5	—	mVpp
TX output resistance	TX_DRIVE = 11b	—	2.0	—	Ω

5.4 Temperature Sensor

The integrated temperature sensor reports the die temperature in TEMP_SENSOR (0x0E). The temperature in degrees Celsius is obtained as °C = value − 40. The register reads 0x00 until the first conversion completes after reset.

Table 5-4. Temperature Sensor Characteristics

Parameter	Min	Typ	Max	Unit
Measurement range	−40	—	+125	°C
Accuracy	−5	—	+5	°C
Resolution	—	1	—	°C
Conversion time	—	1	—	ms

5.5 SPI Interface Timing

SPI mode 0 (CPOL = 0, CPHA = 0) is used, MSB first. The first byte of each transfer is the address byte: bit 7 = R/W (1 = read), bit 6 = auto-increment, bits [5:4] must be 0, bits [3:0] = register address.

Table 5-5. SPI Timing (PVDD = 3.3 V, CL = 20 pF)

Parameter	Symbol	Min	Max	Unit
SCK clock frequency	fSCK	—	10	MHz
SCK high time	tSCKH	40	—	ns
SCK low time	tSCKL	40	—	ns
MOSI setup time	tSU	10	—	ns
MOSI hold time	tH	10	—	ns
NSS setup to first SCK edge	tNSS_SU	50	—	ns
NSS hold after last SCK edge	tNSS_H	50	—	ns
MISO valid after SCK falling edge	tDO	—	25	ns

5.6 I2C Interface Timing

The I2C interface supports Standard-mode and Fast-mode (400 kHz). The 7-bit device address is 0x28 + ADR[1:0], selected by pins ADR1 and ADR0. The device may stretch SCL while internal operations are in progress.

Table 5-6. I2C Timing (Fast-mode)

Parameter	Symbol	Min	Max	Unit
SCL clock frequency	fSCL	—	400	kHz
SCL high time	tHIGH	0.6	—	µs
SCL low time	tLOW	1.3	—	µs
Data setup time	tSU;DAT	100	—	ns
Data hold time	tHD;DAT	0	—	ns
Bus free time between STOP and START	tBUF	1.3	—	µs
Capacitive load per bus line	Cb	—	400	pF

6 Low-Power Card Detection (LPCD)

6.1 Principle of Operation

In LPCD mode the MX-7 remains in a low-power state and periodically emits a short RF pulse to sample the antenna. A card or tag entering the field detunes the antenna and changes the sampled amplitude. When the change exceeds the threshold programmed in LPCD_THR (0x0D), the device sets the LPCD interrupt flag in IRQ_STATUS and asserts IRQ, allowing the host to remain in sleep until a card is actually present.

LPCD is started with the LPCD_START command (0x01). The device first performs a calibration measurement with no card present and then enters the periodic sensing cycle. The cycle is terminated by a detection event, by the IDLE command or by a hardware reset.

6.2 Sensing Period

The sensing period is selected by LPCD_CFG[1:0]. The reset value of LPCD_CFG is 0x01, which selects a default period of 100 ms. Shorter periods reduce detection latency at the expense of a higher average current; longer periods reduce the average current.

Table 6-1. LPCD Period Selection (LPCD_CFG[1:0])

LPCD_CFG[1:0]	Value	Sensing period	Note
00b	0x00	50 ms	
01b	0x01	100 ms	Default after reset
10b	0x02	200 ms	
11b	0x03	400 ms	

LPCD default sensing period: **100 ms** (LPCD_CFG reset value 0x01). LPCD average current at the default period: 12 µA typ (Table 5-2).

6.3 Detection Threshold

LPCD_THR sets the minimum amplitude change, in internal ADC steps, that is interpreted as a card. The reset value 0x20 is suitable for the 40×40 mm reference antenna, giving a typical detection distance of 35 mm. Lower thresholds increase the detection distance but also the probability of false wake-ups caused by metal objects or supply noise. See Errata ERR-01 for operation at low supply voltage.

6.4 Recommended LPCD Sequence

1. Configure RF_CFG and the antenna matching for normal reader operation.
2. Write LPCD_THR (default 0x20) and, if required, LPCD_CFG[1:0].
3. Enable the LPCD interrupt in IRQ_EN and clear pending flags in IRQ_STATUS.
4. Issue LPCD_START (0x01) and put the host into sleep mode.
5. On IRQ, read IRQ_STATUS, clear the LPCD flag, issue RF_ON (0x02) and start card activation.

> LPCD performance depends on antenna geometry, matching and the surrounding metal. Detection distance values in this document are typical values measured with the 40×40 mm reference antenna described in Section 10.

7 Protocol Timing

7.1 Host Timing Requirements

Table 7-1 lists timing parameters relevant for host software. The command response time t_RESP is the time from the end of the host command (NSS rising edge or I2C STOP) until the result is available in the FIFO and the completion interrupt is asserted. The host guard time t_GUARD is an additional margin that the host should add to its software timeouts on top of the FWT computed in Section 7.3.

Table 7-1. Timing Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max
Host guard time (recommended)	t_GUARD	Added to host timeout	0.5 ms	—	—
Command response time, READ BLOCK	t_RESP	TA = 25 °C, Type 2 Tag READ	—	3.5 ms	3.9 ms
Oscillator start-up time	t_OSC	Crystal 27.12 MHz	—	1.2 ms	2.0 ms
Reset pulse width	t_RST	RSTN low	10 µs	—	—
Wake-up time from power-down	t_WAKE	Until registers accessible	—	1.5 ms	2.5 ms
SOFT_RESET completion time	t_SRST	Command to ready	—	0.8 ms	1.0 ms

7.2 Timer Configuration

The timer is started automatically at the end of each transmitted frame when TIMER_CFG[0] (AUTO_START) is set, or manually by the host otherwise. The timer reload value is derived from FWI_CFG. When the timer expires before a valid start-of-frame is detected, the TIMEOUT flag is set in IRQ_STATUS and the receiver is disabled.

7.3 Frame Waiting Time (FWT)

The Frame Waiting Time defines the maximum time the reader waits for a card response after the end of a reader frame. The MX-7 computes FWT from the Frame Waiting time Integer (FWI) stored in FWI_CFG[3:0]:

```
FWT = (256 × 16 / fc) × 2^FWI,   fc = 13.56 MHz
```

The valid FWI range is 0–14; the value 15 is reserved. The default after reset is FWI = 4 (FWI_CFG reset value 0x04), which gives FWT ≈ 4.83 ms (4.833 ms). The largest setting, FWI = 14, gives FWT ≈ 4.949 s. Table 7-2 lists FWT for FWI = 0 to 8.

Table 7-2. Frame Waiting Time vs FWI (fc = 13.56 MHz)

FWI	2^FWI	FWT	Note
0	1	0.302 ms	
1	2	0.604 ms	
2	4	1.208 ms	
3	8	2.417 ms	
4	16	4.833 ms	Default after reset
5	32	9.666 ms	
6	64	19.332 ms	
7	128	38.664 ms	
8	256	77.329 ms	

> One elementary unit 256 × 16 / fc corresponds to 302.065 µs at fc = 13.56 MHz. FWT values in Table 7-2 are rounded to three decimals.

8 Register Map

The MX-7 has sixteen 8-bit registers at addresses 0x00 to 0x0F. Accesses to addresses outside this range are ignored and reads return 0x00. Reserved bits read as 0 and must be written as 0.

Table 8-1. Register Map

Address	Name	Access	Reset value	Description
0x00	CHIP_ID	R	0x71	Device identification. Reads 0x71 on silicon Rev A.
0x01	STATUS	R	0x00	Device state: [0] RF_ON, [1] LPCD_ACTIVE, [2] BUSY, [3] CRC_ERR, [4] PARITY_ERR, [5] COLL.
0x02	CMD	R/W	0x00	Command register. Writing [3:0] starts the command (Section 9).
0x03	FIFO_DATA	R/W	0x00	FIFO data port (64-byte FIFO).
0x04	FIFO_LEVEL	R	0x00	Number of bytes in the FIFO (0–64).
0x05	IRQ_EN	R/W	0x00	Interrupt enable: [0] RX_DONE, [1] TX_DONE, [2] TIMEOUT, [3] FIFO_LOW, [4] FIFO_HIGH, [5] LPCD, [6] ERR.
0x06	IRQ_STATUS	R/W1C	0x00	Interrupt flags (same bit order as IRQ_EN). Write 1 to clear.
0x07	RF_CFG	R/W	0x2C	[1:0] MOD_SEL, [3:2] TX_DRIVE, [5:4] RX_GAIN, [7:6] reserved.
0x08	TX_CFG	R/W	0x10	[2:0] TX bit rate, [4] TX_CRC_EN.
0x09	RX_CFG	R/W	0x22	[1] RX_CRC_EN, [5] RX_PARITY_CHK; other bits reserved.
0x0A	TIMER_CFG	R/W	0x00	[0] AUTO_START, [1] TIMER_EN.
0x0B	FWI_CFG	R/W	0x04	[3:0] FWI (0–14). Default FWI = 4 (Section 7.3).
0x0C	LPCD_CFG	R/W	0x01	[1:0] LPCD period (50/100/200/400 ms). Reset 0x01 = 100 ms.
0x0D	LPCD_THR	R/W	0x20	LPCD detection threshold.
0x0E	TEMP_SENSOR	R	0x00	Die temperature; °C = value − 40.
0x0F	VERSION	R	0x12	Silicon/firmware version code.

8.1 RF_CFG (0x07)

Table 8-2. RF_CFG Bit Fields

Bits	Field	Description
[1:0]	MOD_SEL	00b = ISO/IEC 14443 Type A, 01b = Type B, 1xb = reserved
[3:2]	TX_DRIVE	Transmitter drive strength: 00b = 25 %, 01b = 50 %, 10b = 75 %, 11b = 100 %
[5:4]	RX_GAIN	Receiver gain: 00b = 18 dB, 01b = 23 dB, 10b = 28 dB, 11b = 33 dB
[7:6]	—	Reserved

Reset value 0x2C: MOD_SEL = 00b (Type A), TX_DRIVE = 11b (100 %), RX_GAIN = 10b (28 dB).

8.2 FWI_CFG (0x0B)

FWI_CFG[3:0] holds the Frame Waiting time Integer used by the timer unit. The reset value 0x04 selects FWI = 4. Bits [7:4] are reserved. Writing 15 (0xF) to FWI_CFG[3:0] is not allowed.

8.3 LPCD_CFG (0x0C)

Table 8-3. LPCD_CFG Bit Fields

Bits	Field	Description
[1:0]	PERIOD	00b = 50 ms, 01b = 100 ms, 10b = 200 ms, 11b = 400 ms
[3:2]	—	Reserved
[4]	AUTO_RF_ON	1 = switch the RF field on automatically after a detection event
[7:5]	—	Reserved

Reset value 0x01 (PERIOD = 01b, 100 ms; AUTO_RF_ON = 0).

8.4 TEMP_SENSOR (0x0E)

Read-only. The register is updated once per conversion (Table 5-4). Temperature: °C = value − 40. Example: a reading of 0x41 (65) corresponds to 25 °C.

8.5 IRQ_STATUS (0x06)

Each flag is set by hardware when the corresponding event occurs, regardless of IRQ_EN. The IRQ pin is asserted when any flag is set whose enable bit in IRQ_EN is 1. Flags are cleared by writing 1 to the bit position (R/W1C).

9 Command Set

Commands are started by writing the command code to CMD[3:0]. While a command is executing, STATUS[2] (BUSY) is set. Writing IDLE terminates any running command.

Table 9-1. Command Set

Code	Command	Description
0x00	IDLE	Stop the current command and return to idle
0x01	LPCD_START	Start Low-Power Card Detection (Section 6)
0x02	RF_ON	Switch the RF carrier on
0x03	RF_OFF	Switch the RF carrier off
0x0C	TRANSCEIVE	Transmit FIFO content, then receive the card response into the FIFO
0x0F	SOFT_RESET	Reset all registers to their reset values

All other command codes are reserved. TRANSCEIVE starts the FWT timer automatically when TIMER_CFG[0] is set. After SOFT_RESET the host should wait for t_SRST (Table 7-1) before accessing the device.

10 Application Information

10.1 Typical Application

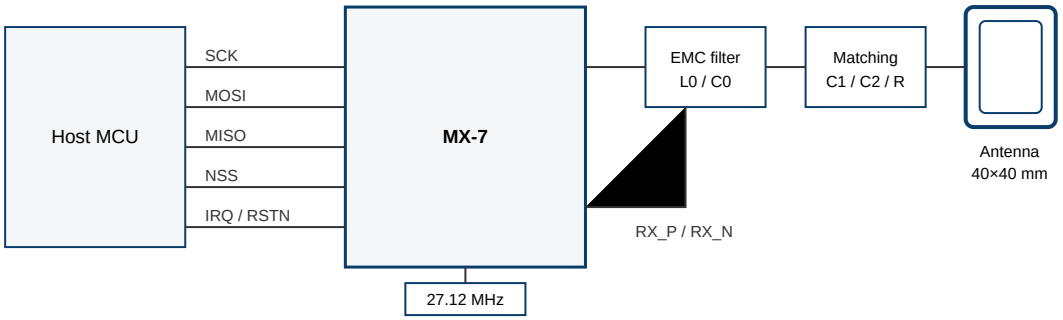


Figure 10-1. Typical Application with SPI Host

Figure 10-1 shows a typical application. The TX outputs drive the antenna through a second-order EMC filter (L0/C0) and a matching network (C1/C2 with damping resistor R). The receiver is coupled to the antenna through a resistive divider and coupling capacitors. Table 10-1 gives starting values for the 40×40 mm reference antenna.

Table 10-1. Matching Network Starting Values (40×40 mm reference antenna)

Component	Value	Note
L0	560 nH	EMC filter inductor, ±5 %
C0	220 pF	EMC filter capacitor, C0G
C1	68 pF	Series matching capacitor, C0G
C2	150 pF	Parallel matching capacitor, C0G
R	1.5 Ω	Damping resistor (quality factor adjustment)
Antenna	40×40 mm, 4 turns	Inductance approx. 1.2 µH

10.2 Power Supply Decoupling

- Place a 10 µF and a 100 nF capacitor at TVDD, as close as possible to pins 8 and 10.
- Decouple AVDD, DVDD and VMID with 100 nF each; VREG with 1 µF.
- Connect TVSS1, TVSS2, AVSS, DVSS and the exposed pad to a solid ground plane with multiple vias.

10.3 Layout Guidelines

- Keep the TX traces short and symmetrical; route TX1 and TX2 as a differential pair.
- Keep the crystal and its load capacitors close to XTAL1/XTAL2 and away from the TX traces.
- Avoid routing digital host signals underneath the antenna matching network.
- Provide at least 5 mm clearance between the antenna and any metal; use ferrite foil if metal is unavoidable.

10.4 Host Software Initialization

1. Apply VDD and PVDD, release RSTN and wait at least t_OSC (Table 7-1).
2. Read CHIP_ID (expected 0x71) and VERSION (expected 0x12).
3. Write RF_CFG, TX_CFG and RX_CFG for the selected protocol.
4. Write FWI_CFG with the FWI required by the card, compute FWT per Section 7.3 and set the host software timeout to FWT plus t_GUARD (Table 7-1).
5. Issue RF_ON and start the card activation sequence.

10.5 NFC Forum Type 2 Tag Access

To read a Type 2 Tag, the host writes the READ command (0x30) followed by the block number into the FIFO and issues TRANSCEIVE. The tag returns 16 bytes (four blocks) followed by CRC_A. A WRITE (0xA2) transfers 4 bytes to a single block and is acknowledged by a 4-bit ACK. Typical READ BLOCK response time is given in Table 7-1.

11 Package Information

11.1 QFN-32 5x5 mm Package Dimensions

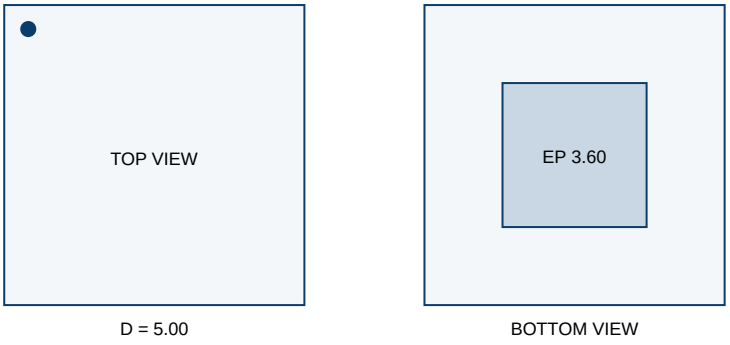


Figure 11-1. QFN-32 5x5 mm Outline (dimensions in mm)

Table 11-1. QFN-32 5x5 mm Package Dimensions (mm)

Symbol	Description	Min	Nom	Max
A	Total height	0.80	0.85	0.90
A1	Standoff	0.00	0.02	0.05
D / E	Body size	4.90	5.00	5.10
D2 / E2	Exposed pad size	3.45	3.60	3.75
e	Lead pitch	—	0.50 BSC	—
b	Lead width	0.18	0.25	0.30
L	Lead length	0.30	0.40	0.50

11.2 Recommended Land Pattern

Use a land pattern per IPC-7351 for 0.50 mm pitch QFN. The exposed pad land should be 3.60 mm × 3.60 mm with a 3×3 array of thermal vias (0.3 mm drill) connected to the ground plane. A segmented solder paste stencil with approximately 60 % paste coverage on the exposed pad is recommended.

11.3 Package Marking

Table 11-2. Package Marking

Line	Content	Example
1	Device name	MX7
2	Silicon revision + date code (YYWW)	A2508
3	Assembly lot code	L00123

11.4 Packing

Devices are delivered on tape and reel, 4000 units per 13-inch reel, pin 1 oriented toward the sprocket holes. Moisture-sensitive handling per MSL 3 applies (Table 4-3).

12 Errata

This section lists known deviations of the MX-7 from the behavior specified in this document. Errata are identified by ERR-nn numbers that remain stable across document revisions.

Table 12-1. Errata Summary

ID	Title	Affected silicon	Status
ERR-01	LPCD false wake-up at low VDD	Rev A	Open
ERR-02	I2C clock stretching after SOFT_RESET	Rev A	Open

ERR-01 — LPCD may wake falsely when VDD < 2.8 V

Open

Description: LPCD may wake falsely when VDD < 2.8 V. At supply voltages below 2.8 V the calibration amplitude measured at LPCD_START is reduced, and supply noise can exceed the default threshold.

Workaround: raise LPCD_THR. A value of 0x30 or higher is recommended when VDD can drop below 2.8 V.

ERR-02 — I2C clock stretching may exceed 1 ms after SOFT_RESET

Open

Description: I2C clock stretching may exceed 1 ms after SOFT_RESET. Hosts with a short bus timeout may report a bus error.

Workaround: after SOFT_RESET, wait at least 2 ms before the next I2C access, or use the SPI interface.

Appendix A — Abbreviations

Table A-1. Abbreviations

Term	Meaning
ASK	Amplitude shift keying
CRC	Cyclic redundancy check
EP	Exposed pad
FIFO	First-in, first-out buffer
FWI	Frame Waiting time Integer
FWT	Frame Waiting Time
HF	High frequency (13.56 MHz)
LPCD	Low-Power Card Detection
NFC	Near Field Communication
QFN	Quad flat no-lead package
RX / TX	Receive / transmit
Tj / TA	Junction / ambient temperature

Revision History

Table R-1. Revision History

Version	Date	Silicon	Description
1.0	2024-06	—	Preliminary release.
1.1	2024-10	—	Added LPCD characterization data; corrected pin 22 description.
1.2	2025-03	A	Production data for silicon Rev A. Added Errata ERR-01 and ERR-02.

Nurichips (fictional) · MX-7 Datasheet v1.2. This document and the company and product it describes are fictional and were created solely for training purposes. No warranty of any kind is given. ISO/IEC 14443 and NFC Forum are referenced by name only.